

```
# @lit-labs/ssr-dom-shim
```

```
## Overview
```

This package provides minimal implementations of ``Element``, ``HTMLElement``, ``EventTarget``, ``Event``, ``CustomEvent``, ``CustomElementRegistry``, and ``customElements``, designed to be used when Server Side Rendering (SSR) web components from Node, including Lit components.

```
## Usage
```

```
### Usage from Lit
```

Lit itself automatically imports these shims when running in Node, so Lit users should typically not need to directly depend on or import from this package.

See the [lit.dev SSR docs](https://lit.dev/docs/ssr/overview/) for general information about server-side rendering with Lit.

```
### Usage in other contexts
```

Other libraries or frameworks who wish to support SSR are welcome to also depend on these shims. (This package is planned to eventually move to `@webcomponents/ssr-dom-shim`` to better reflect this use case). There are two main patterns for providing access to these shims to users:

1. Assigning shims to ``globalThis``, ensuring that assignment occurs before user-code runs.
2. Importing shims directly from the module that provides your base class, using the ``node`` [export condition](https://nodejs.org/api/packages.html#conditional-exports) to ensure this only happens when running in Node, and not in the browser.

Lit takes approach #2 for all of the shims except for ``customElements``, ``Event`` and ``CustomEvent``, so that users who have imported ``lit`` are able to call ``customElements.define`` or ``new Event(...)``/``new CustomEvent(...)`` in their components from Node.

```
### Exports
```

The main module exports the following values. Note that no globals are set by this module.

```
- [ `EventTarget` ](https://developer.mozilla.org/en-US/docs/Web/API/EventTarget)
- [ `addEventListener` ](https://developer.mozilla.org/en-US/docs/Web/API/EventTarget/addEventListener)
- [ `dispatchEvent` ](https://developer.mozilla.org/en-US/docs/Web/API/EventTarget/dispatchEvent)
- [ `removeEventListener` ](https://developer.mozilla.org/en-US/docs/Web/API/EventTarget/removeEventListener)
- [ `Element` ](https://developer.mozilla.org/en-US/docs/Web/API/Element)
- (Inherits from EventTarget)
- [ `attachShadow` ](https://developer.mozilla.org/en-US/docs/Web/API/Element/attachShadow)
- [ `shadowRoot` ](https://developer.mozilla.org/en-US/docs/Web/API/Element/shadowRoot)
- [ `attributes` ](https://developer.mozilla.org/en-US/docs/Web/API/Element/attributes)
- [ `hasAttribute` ](https://developer.mozilla.org/en-US/docs/Web/API/Element/hasAttribute)
- [ `getAttribute` ](https://developer.mozilla.org/en-US/docs/Web/API/Element/getAttribute)
- [ `setAttribute` ](https://developer.mozilla.org/en-US/docs/Web/API/Element/setAttribute)
- [ `removeAttribute` ](https://developer.mozilla.org/en-US/docs/Web/API/Element/removeAttribute)
- [ `HTMLElement` ](https://developer.mozilla.org/en-US/docs/Web/API/HTMLElement)
- (Inherits from Element)
- [ `CustomElementRegistry` ](https://developer.mozilla.org/en-US/docs/Web/API/CustomElementRegistry)
- [ `customElements` ](https://developer.mozilla.org/en-US/docs/Web/API/Window/customElements)
- [ `Event` ](https://developer.mozilla.org/en-US/docs/Web/API/Event)
- [ `CustomEvent` ](https://developer.mozilla.org/en-US/docs/Web/API/CustomEvent)
- [ `MediaList` ](https://developer.mozilla.org/en-US/docs/Web/API/MediaList)
- [ `StyleSheet` ](https://developer.mozilla.org/en-US/docs/Web/API/StyleSheet)
- [ `CSSRule` ](https://developer.mozilla.org/en-US/docs/Web/API/CSSRule)
- [ `CSSRuleList` ](https://developer.mozilla.org/en-US/docs/Web/API/CSSRuleList)
- [ `CSSStyleSheet` ](https://developer.mozilla.org/en-US/docs/Web/API/CSSStyleSheet)
- (Inherits from StyleSheet)
- [ `replace` ](https://developer.mozilla.org/en-US/docs/Web/API/CSSStyleSheet/replace)
- [ `replaceSync` ](https://developer.mozilla.org/en-US/docs/Web/API/CSSStyleSheet/replaceSync)
```

```
### CSS Node.js customization hook
```

`@lit-labs/ssr-dom-shim/register-css-hook.js`` implements/registers a [Node.js customization hook](https://nodejs.org/api/module.html#customization-hooks) (Node.js >= 18.6.0) to import CSS files/modules as instances of ``CSSStyleSheet``.

```
```ts
import styles from 'my-styles.css' with {type: 'css'};
// styles is now an instance of CSSStyleSheet
```
```

This can either be used as a parameter with the Node.js CLI (e.g. ``node --import @lit-labs/ssr-dom-shim/register-css-hook.js my-script.js`` or via environment variable ``NODE_OPTIONS="--import @lit-labs/ssr-dom-shim/register-css-hook.js"``) or imported inline, and it will apply to any module dynamically imported afterwards

(e.g. ``import @lit-labs/ssr-dom-shim/register-css-hook.js`` and subsequently ``await import('./my-component.js')``).

- [Node.js Customization Hooks](https://nodejs.org/api/module.html#customization-hooks)
- [Import Attributes](https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Statements/import/with)

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chromeos_smart_card_connector

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```

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*
* Optimised ANSI C code for the Rijndael cipher (now AES)
*
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wcwidth.js modifications

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wcwidth.c

Markus Kuhn -- 2007-05-26 (Unicode 5.0)

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